IIMT3642: Managing and Mining Big Data - Classification (Decision Tree I)

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Lecture Outline

- What is classification?
- An impurity measure for a dataset

Readings for this lecture

- **Data Science for Business**, Foster Provost and Tom Fawcett, O'Reilly, 2013.
  - pp. 43 - 56

What is classification?

Examples of classification questions to ask during predictive data analysis

- Retail: whether a customer is expecting a baby
- Telecom: whether a customer will stop using ours and churn to competitors
- Banking: whether a customer will default her/his loan
- Target Marketing: What a customer's response to a catalog will be (e.g., no response, high profitable purchase, low profitable purchase)

What is the commonality among these examples?

- assign an instance of interest to one of the predefined classes

1. Retail: whether a customer is expecting a baby → {expecting a baby, NOT expecting a baby}
2. Telecom: whether a customer will churn → {churn, NOT churn}
3. Banking: whether a customer will default her/his loan →{default, NOT default}
4. Target Marketing: predict a customer's response to a catalog → {no response, purchase high profitable products, purchase low profitable products}

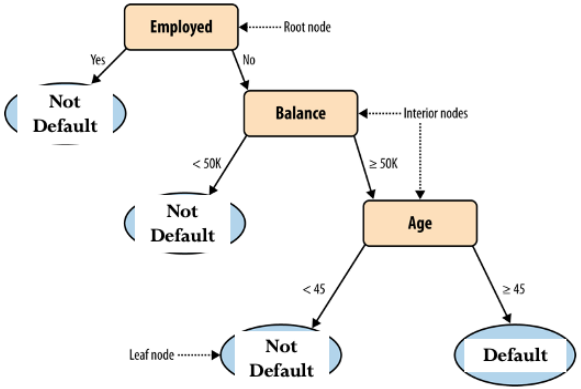
- Classification is the process of learning a model (from training data) that can assign an instance of interest (whose class is unknown) to one of the predefined classes.
  - A classification model is a predictive model because the class of an instance to be classified is unknown.
  - For example, we want to predict whether a customer belongs to the “Churn” class or the “Not Churn” class.

- Training data consist of instances whose class is known. Each instance is described with descriptive attributes and a class label attribute.

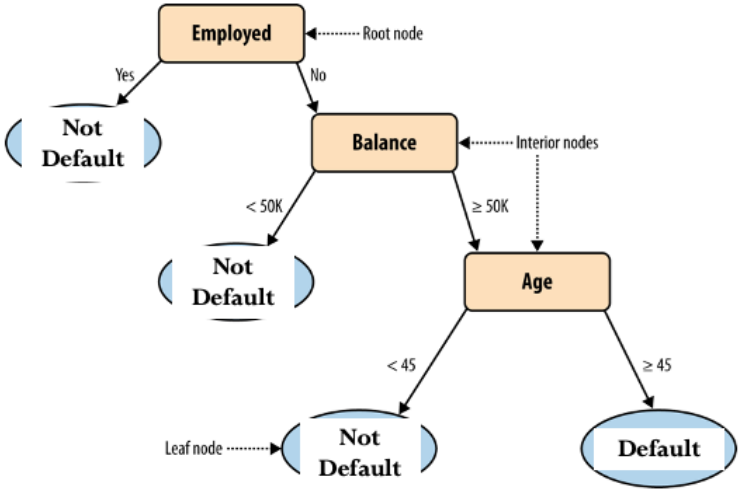
Name	Balance	Age	Employed	Default
Mike	$200,000	42	No	Yes
Mary	$350,00	33	Yes	No
Claudio	$115,000	40	No	No
Robert	$29,000	23	Yes	Yes
Dora	$72,000	31	No	No

The classification process consists of two steps:

- Step 1: Learn a classification model from training data;
- Step 2: Apply the model to classify an instance with unknown class.



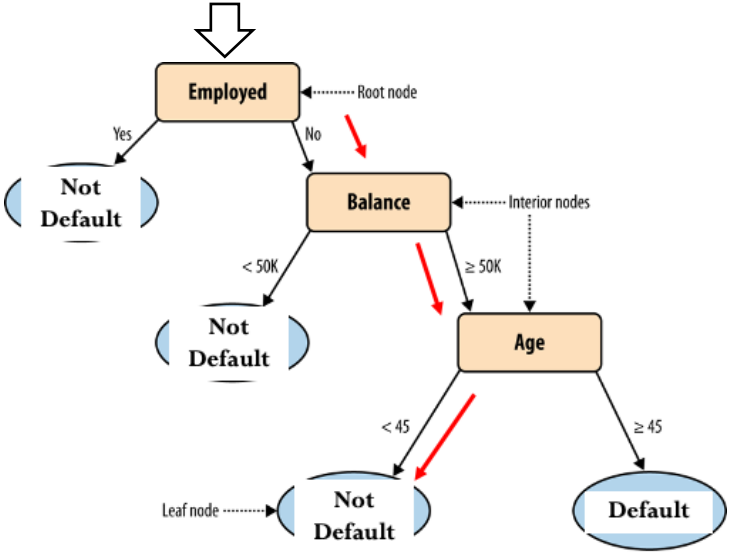
- A decision tree model is learned from training data and represents decisions as an upside-down tree, where each internal node tests a feature and each branch leads to an outcome.

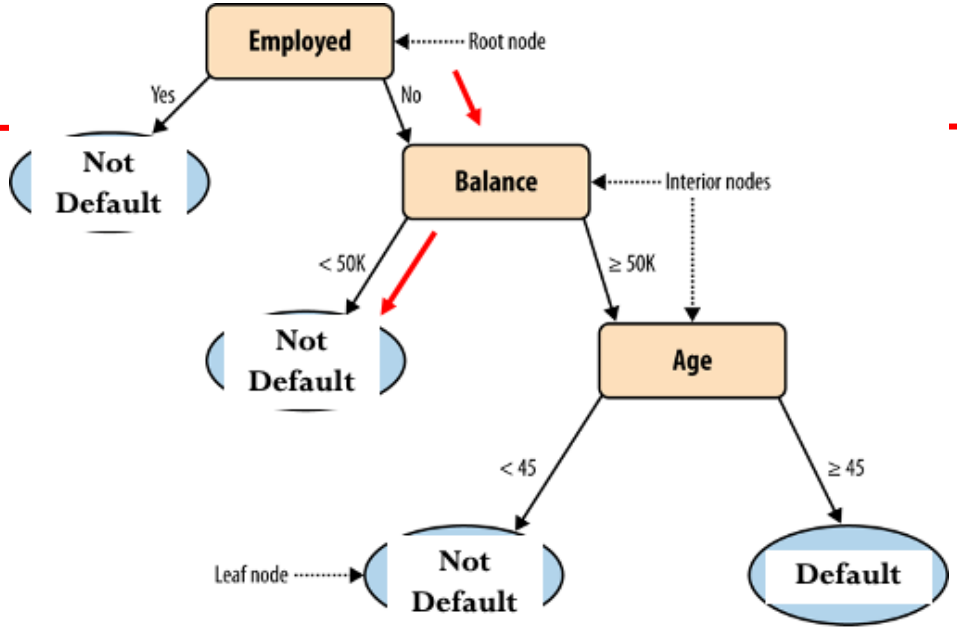


- Each internal node or the root node represents a test of a descriptive attribute and the test result determines which branch to go.
- Eventually, the sequence of decisions reaches a terminal leaf node, which represents a predicted class.

When testing/predicting an unknown instance

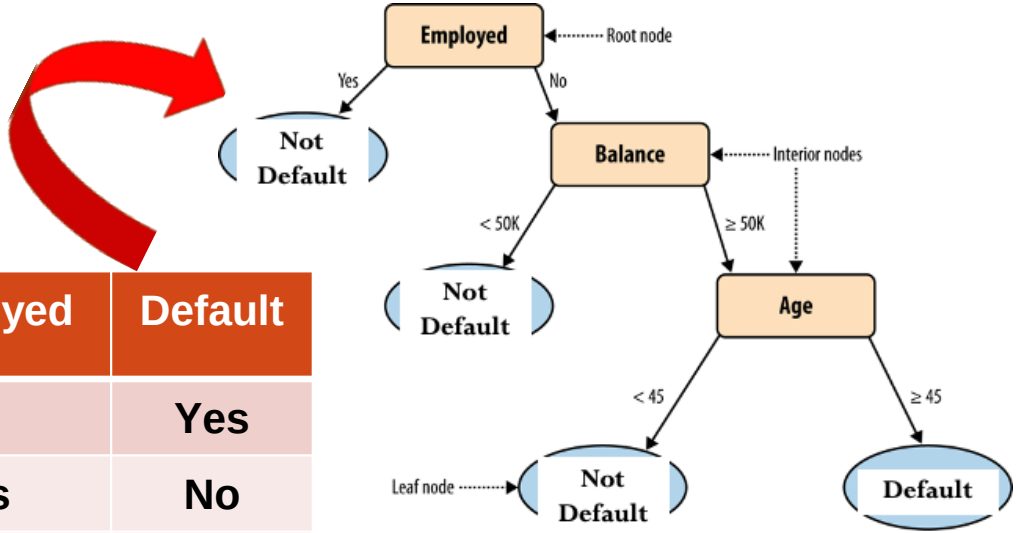
Name	Balance	Age	Employed	Default
Jay	$120,000	42	No	?





Name	Balance	Age	Employed	Default
Leon	$20,000	31	No	?

How did we train for this tree-model in the first place, given training data?



Name	Balance	Age	Employed	Default
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Robert	$29,000	23	Yes	Yes
Dora	$72,000	31	No	No

An Impurity Measure

General idea of learning a decision tree model: find attributes to divide training data and make resulting subsets of training data as pure as possible.

- Purity/impurity is defined based on the distribution of the class label attribute.
- Measure of purity/impurity?

Name	Balance	Age	Employed	Default
				Yes
				No
				No
				Yes
				No

Name	Balance	Age	Employed	Default
				No
				No
				No
				No
				No

Which data set is more PURE?

Entropy is an impurity measure for a data set.

- The higher the value of entropy the more impure a data set is.
- Basically, high entropy value is BAD.

Entropy of a data set:

$$Entropy = -p_1 \times \log_2 p_1 - p_2 \times \log_2 p_2 - \cdots - p_k \times \log_2 p_k$$

k : number of predefined classes

p_i : fraction of instances belonging to the i-th class

Note that $0 \times \log_2 0 = 0$ by convention

(optional: because of L'Hopital's rule $\lim_{x \to 0^+} x \log(x) = 0$)

Class Probabilistic Distribution:

$$p_{NO} = 1$$
$$p_{YES} = 0$$

$$\begin{aligned} Entropy &= -\,pNO \times \log_2 p_{NO} - p_{YES} \times \log_2 p_{YES} \\ &= -1 \times \log_2 1 - 0 \times \log_2 0 \\ &= 0 \end{aligned}$$

Name	Balance	Age	Employed	Default
				No
				No
				No
				No
				No
				No
				No
				No
				No
				No

Class Probabilistic Distribution

$$p_{NO} = 0$$
$$p_{YES} = 1$$

$$\begin{aligned} Entropy &= -p_{NO} \times \log_2 p_{NO} - p_{YES} \times \log_2 p_{YES} \\ &= -0 \times \log_2 0 - 1 \times \log_2 1 \\ &= 0 \end{aligned}$$

The smallest entropy value is 0, indicating all instances in a data set belong to one class.

Name	Balance	Age	Employed	Default
				Yes
				Yes
				Yes
				Yes
				Yes
				Yes
				Yes
				Yes
				Yes
				Yes

Class Probabilistic Distribution

$$p_{NO} = 0.8$$
$$p_{YES} = 0.2$$

$$\begin{aligned} Entropy &= -\,p_{NO} \times \log_2 p_{NO} - p_{YES} \times \log_2 p_{YES} \\ &= -0.8 \times \log_2 0.8 - 0.2 \times \log_2 0.2 \\ &= 0.72 \end{aligned}$$

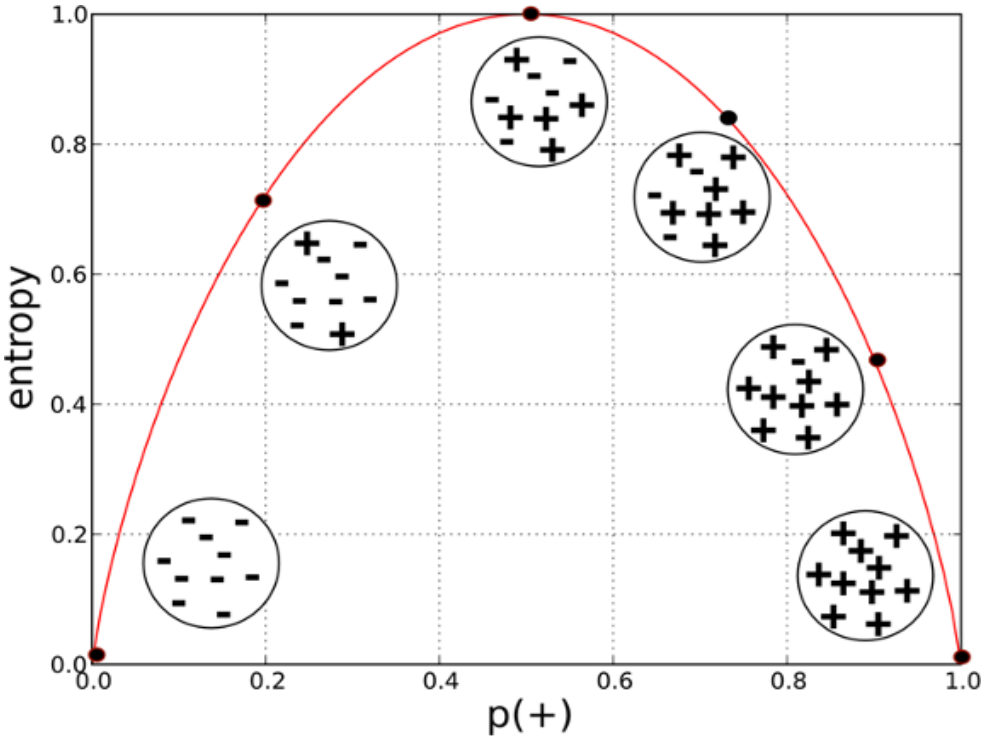
Name	Balance	Age	Employed	Default
				No
				No
				No
				No
				Yes
				No
				No
				No
				Yes
				No

Class Probabilistic Distribution

$$p_{NO} = 0.5$$
$$p_{YES} = 0.5$$

$$\begin{aligned} Entropy &= -\,p_{NO} \times \log_2 p_{NO} - p_{YES} \times \log_2 p_{YES} \\ &= -0.5 \times \log_2 0.5 - 0.5 \times \log_2 0.5 \\ &= 1 \end{aligned}$$

Name	Balance	Age	Employed	Default
				No
				Yes
				No
				Yes
				Yes
				No
				Yes
				No
				Yes
				No



+ : Default
– : Not Default

What if there are more than 2 classes?

$p_{NO} = 1$

$p_{high\ profit} = 0$

$p_{low\ profit} = 0$

$$Entropy = -\ pNO \times \log_2 pN_{O\ _}\ p_{high\ profit} \times \log_2 p_{high\ profit} -\ plow_{\ profit} \times \log_2 pl_{ow\ profit} = 0$$

				Response
				No
				No
				No
				No
				No
				No
				No
				No
				No
				No

$p_{NO} = 0$

$p_{high\ profit} = 1$

$p_{low\ profit} = 0$

$$Entropy = -\ pNO \times \log_2 pN_{O\ _}\ p_{high\ profit} \times \log_2 p_{high\ profit} -\ plow_{\ profit} \times \log_2 pl_{ow\ profit} = 0$$

				Response
				High Profit
				High Profit
				High Profit
				High Profit
				High Profit
				High Profit
				High Profit
				High Profit
				High Profit
				High Profit

$p_{NO} = 0.2$

$p_{high\ profit} = 0.2$

$p_{low\ profit} = 0.6$

$$Entropy = -\ pNO \times \log_2 pN_{O\ _}\ p_{high\ profit} \times \log_2 p_{high\ profit} -\ plow_{\ profit} \times \log_2 pl_{ow\ profit} = 1.37$$

Quick fact: When the # of classes is greater than 2, entropy value could go beyond 1.

				Response
				High Profit
				Low Profit
				High Profit
				High Profit
				Low Profit
				No
				High Profit
				No
				High Profit
				High Profit